

If you're running models locally, thinking "bigger memory pool = better AI box" falls apart the moment you care about actual speed.

Capacity decides whether the model fits.

Bandwidth decides whether the box feels alive or like it's decoding through wet cement to produce 3 tokens per second.

That is why a 32GB RTX 5090 and RTX PRO 6000 can absolutely outrun a much larger unified-memory machine, while a Mac Studio M3 Ultra, DGX Spark, or Strix Halo box can still be the right answer when the model simply will not fit on a normal GPU (but will be a lot slower, and not help for multi-agentic workflows).

There's a better way to think about it:

Local AI hardware = capacity × bandwidth × software stack

Capacity tells you what fits.

Bandwidth tells you how hard the box can breathe.

The software stack tells you how much of the spec sheet you can actually cash out.

That is the mental model.

Not "AI PC."

Not "NPU TOPS."

Not whatever crime against engineering was committed in a marketing deck this week.

The Hardware Number You Should Actually Care About

Especially in the age of agents.

Memory bandwidth is not tokens per second.

But it is the cleanest first-pass way to separate local AI hardware into real performance tiers before you waste a week arguing with someone posting screenshots from a single prompt demo.

Here's the current landscape:

1.8 TB/s class

- RTX PRO 6000 Blackwell, RTX 5090 → **1792 GB/s**

800 GB/s class

- Mac Studio M3 Ultra → **819 GB/s**

450–650 GB/s class

- Mac Studio M4 Max → **546 GB/s**
- MacBook Pro M5 Max → **460–614 GB/s**
- AMD Radeon AI PRO R9700 → **640 GB/s**
- Tenstorrent Blackhole p150 → **512 GB/s**

250–300 GB/s unified-memory class

- DGX Spark → **273 GB/s**
- Mac mini M4 Pro → **273 GB/s**
- Ryzen AI Max / Strix Halo → **256 GB/s**

Thin-and-light AI PC class

- MacBook Air M5 → **153 GB/s**
- Snapdragon X Elite → **135 GB/s**
- Intel Lunar Lake → **136 GB/s**
- Snapdragon X2 Elite → **152–228 GB/s**

If you remember nothing else, remember this:

- capacity decides what fits
- bandwidth decides how hard it can breathe
- software decides how much of that you actually see

That is the whole game.

Side Note: The Memory Tax People Mix Up

A lot of people collapse *capacity* and *bandwidth* into one blob.

That is how you end up making terrible hardware takes with supreme confidence.

A 32GB RTX 5090 and a 96GB RTX PRO 6000 Blackwell both have the same bandwidth.

But they live in completely different worlds once model size enters the chat.

A DGX Spark gives you 128GB unified memory at 273 GB/s.

A Ryzen AI Max system can expose ~96GB as GPU memory.

A Mac Studio M3 Ultra goes up to 512GB at 819 GB/s.

Same topic.

Wildly different tradeoffs.

So no, bandwidth is not the whole story.

But it is the fastest way to stop being confused.

What This Looks Like in Practice

Below ~150 GB/s, you are in thin-and-light territory.

That does not mean useless.

It means stop pretending it's competing with workstation GPUs.

Around **250–300 GB/s** → unified memory starts getting interesting.

Around **450–650 GB/s** → serious workstation tier.

At **800+ GB/s** → expensive, powerful, and fun.

Local AI in 2026 is not one market.

It is five different markets pretending to be one.

Discrete GPU Reality: Still the Bandwidth Kings

If the model fits, or you pool GPUs via NVLink (now mostly server-side) or Gen 5 PCIe and use Tensor Parallelism, discrete GPUs still dominate. This applies especially to NVIDIA GPUs given the wide software support.

- RTX PRO 6000 Blackwell → 96GB @ 1792 GB/s
- RTX 5090 → 32GB @ 1792 GB/s
- RTX 4090 → 24GB @ 1008 GB/s

What about AMD GPUs?

- RX 7900 XTX → 24GB @ 960 GB/s
- Radeon PRO W7900 → 48GB @ 864 GB/s
- AI PRO R9700 → 32GB @ 640 GB/s

Intel?

- Arc Pro B65 → 32GB @ ~608 GB/s
- Arc Pro B60 → 24GB @ ~456 GB/s

GPUs win because they can drink from a firehose.

They lose when the model doesn't fit.

Apple Reality: OK Bandwidth + Capacity Together

Apple's entire story is:

not the fastest, but usable

- Mac mini M4 → 120 GB/s
- MacBook Air M5 → 153 GB/s
- Mac mini M4 Pro → 273 GB/s
- MacBook Pro M5 Pro → 307 GB/s

- M5 Max → up to 614 GB/s
- Mac Studio M3 Ultra → **819 GB/s + up to 512GB memory**

That last one is the key.

Apple wins when:

- you want one box
- you want silence
- you want stupid amounts of memory
- you don't want to shard across GPUs

It loses when raw tokens/sec & concurrency matter more than everything else.

DGX Spark: Coherent Memory + CUDA, Not a Bandwidth Monster

DGX Spark:

- 128GB unified memory
- 273 GB/s
- NVIDIA stack

That bandwidth is not impressive.

The **coherent memory + software stack** is.

It's a developer appliance.

Not a raw performance monster. It has NVFP4 support, which gives it an advantage, but that is yet to mature.

Strix Halo / Ryzen AI Max: The First Real x86 Contender

This is an interesting one.

- 256-bit LPDDR5X
- up to 128GB memory
- ~256 GB/s bandwidth
- up to ~96GB usable as GPU memory

This is also where Framework Desktop is interesting.

The AI PC Trap

Most "AI PCs" are still bandwidth-starved.

- Snapdragon X Elite → 135 GB/s
- Intel Lunar Lake → 136 GB/s
- MacBook Air M5 → 153 GB/s
- Snapdragon X2 Elite → up to ~228 GB/s

That's fine for:

- small models
- assistants
- edge workloads

It is not:

- 9B Dense model playground territory
- serious multi-agent workloads
- long-context stress testing

Physics still applies.

Tenstorrent and the Wildcards

Tenstorrent:

- Wormhole n300 → 24GB @ 576 GB/s
- Blackhole p150 → 32GB @ 512 GB/s + 800G interconnect

Fully OSS stack. I am excited or this one to mature and hope they go on to become a strong contender in the AI space. We need more full opensource stacks.

They are real options depending on your stack and goals.

Why Bigger Boxes Still Feel Slow

Because fitting $\neq$ serving.

Even if it fits, you still pay for:

- bandwidth during decode
- KV cache growth
- dequantization
- batching + concurrency
- scheduler quality
- framework overhead

This is why:

“it runs” = demo “it serves” = system design

Multi-GPU?

More GPUs $\neq$ linear scaling.

You are now buying:

- interconnect (PCIe vs NVLink vs RDMA)
- topology
- sync overhead
- software maturity

The Only Mental Model That Matters

There is no giant chart you need to memorize.

There is just this:

1. What must fit?

2. What bandwidth tier do I need?

3. What software stack can actually deliver it?

Blunt version:

- NVIDIA $\rightarrow$ fastest raw speed
- Apple Ultra $\rightarrow$ biggest one-box memory
- Strix Halo $\rightarrow$ first real x86 unified-memory play
- DGX Spark $\rightarrow$ coherent NVIDIA appliance
- AMD / Intel Arc $\rightarrow$ rising alternatives
- Tenstorrent $\rightarrow$ fully opensource stack

Once you internalize this, you stop asking:

“Which hardware is best?”

You start asking:

“Which bottleneck am I buying?”

That is the real question.